Gaussian Mixture Model (GMM) Using Expectation-Maximization (EM) Algorithm

Project Report

Submitted By

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1. Introduction

Machine Learning provides powerful techniques for discovering hidden patterns within data. One of the most widely used unsupervised learning techniques is the Gaussian Mixture Model (GMM), which represents data as a combination of multiple Gaussian distributions.

In this project, a Gaussian Mixture Model was implemented from scratch using the Expectation-Maximization (EM) Algorithm in R. The implementation does not use any external machine learning libraries and demonstrates the mathematical foundation behind probabilistic clustering.

The project generates a synthetic dataset, estimates model parameters iteratively, performs clustering, evaluates convergence using log-likelihood, and visualizes the final clustering results.

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2. Problem Statement

The objective of this project is to cluster unlabeled data points into meaningful groups using a probabilistic approach.

Traditional clustering methods such as K-Means assign each data point to only one cluster. In contrast, GMM assigns probabilities indicating the likelihood that a data point belongs to each cluster.

The challenge is to estimate:

- Cluster membership probabilities
- Mean vectors
- Covariance matrices
- Mixing coefficients

without knowing the true cluster labels.

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3. Objectives

The primary objectives of this project are:

- Generate synthetic Gaussian-distributed datasets.
- Implement Gaussian Mixture Model manually.
- Apply Expectation-Maximization Algorithm.

- Estimate cluster parameters.
- Perform probabilistic clustering.
- Monitor model convergence.
- Visualize clustering results.
- Analyze model performance on large datasets.

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4. Dataset Generation

A synthetic dataset was generated using two Gaussian distributions.

Dataset Characteristics

Parameter	**Value**
Total Records	20,000
Features	2
Clusters	2
Data Type	Synthetic

Cluster 1

Mean Vector:

$\mu_1 = [2, 3]$

Covariance Matrix:

[1.0 0.5]

[0.5 1.0]

Cluster 2

Mean Vector:

$\mu_2 = [7, 8]$

Covariance Matrix:

[1.0 -0.3]

[-0.3 1.0]

The dataset was generated using Cholesky decomposition to transform standard normal random variables into correlated Gaussian samples.

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5. Methodology

The project follows the Gaussian Mixture Model framework combined with the Expectation-Maximization algorithm.

Step 1: Initialization

The following parameters were initialized:

Mixing Coefficients

$\pi_k = 1/K$

Mean Vectors

Random data points were selected as initial cluster centers.

Covariance Matrices

Identity matrices were used as initial covariance matrices.

Step 2: Gaussian Density Computation

The Gaussian Probability Density Function was implemented manually.

Formula:

$N(x|\mu,\Sigma)$

This function calculates the probability density of each observation under a specific Gaussian distribution.

Step 3: Expectation Step (E-Step)

Responsibilities were calculated for each data point.

Responsibilities represent the probability that a point belongs to a particular cluster.

The output of the E-Step is a responsibility matrix γ.

Step 4: Maximization Step (M-Step)

Using the responsibility matrix:

- Mixing coefficients were updated.
- Mean vectors were updated.
- Covariance matrices were updated.

These updates maximize the expected log-likelihood of the observed data.

Step 5: Log-Likelihood Calculation

The overall likelihood of the model was calculated after each iteration.

Purpose:

- Evaluate model quality.
- Track convergence.
- Determine when parameter updates become stable.

Step 6: Convergence Check

The EM algorithm was repeated until:

- Maximum iterations were reached, or
- Log-likelihood difference became smaller than 0.0001.

This indicates that the model parameters have stabilized.

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6. Implementation Details

Programming Language:

R

Key Components Implemented:

- Dataset Generator
- Parameter Initialization
- Gaussian Density Function
- E-Step Function
- M-Step Function
- Log-Likelihood Function
- EM Iteration Loop
- Cluster Assignment
- Visualization Module

No external machine learning libraries were used.

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7. Output Generated

The project automatically generates the following outputs:

Dataset File

synthetic_data_20000.csv

Cluster Output

GMM_Cluster_Output.csv

Dataset Visualization

Dataset.png

Final Cluster Visualization

Final_Clusters.png

Log-Likelihood Plot

LogLikelihood.png

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8. Results

The Gaussian Mixture Model successfully identified the underlying Gaussian clusters within the dataset.

Observations:

- Mixing coefficients converged close to 0.5.
- Cluster centers converged near the original means.
- Log-likelihood increased monotonically.
- Final clusters were clearly separated.
- The EM algorithm successfully recovered hidden structure from the data.

The clustering results closely matched the true distribution used to generate the synthetic dataset.

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9. Advantages of GMM

- Supports soft clustering.
- Models complex cluster shapes.
- Provides probabilistic interpretation.
- More flexible than K-Means.
- Useful for density estimation.

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10. Limitations

- Computationally expensive for large datasets.
- Sensitive to initialization.
- May converge to local optima.
- Requires covariance matrix inversion.

For very large datasets, optimized matrix-based implementations are recommended.

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11. Applications

Gaussian Mixture Models are widely used in:

- Customer Segmentation
- Market Analysis
- Image Segmentation
- Speech Recognition
- Medical Data Analysis
- Pattern Recognition
- Anomaly Detection
- Bioinformatics

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12. Learning Outcomes

Through this project, the following concepts were learned:

- Unsupervised Machine Learning
- Gaussian Distributions
- Probability Density Functions
- Covariance Matrices
- Expectation-Maximization Algorithm
- Density Estimation
- Probabilistic Clustering
- Statistical Data Analysis
- Data Visualization in R

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13. Conclusion

This project successfully implemented a Gaussian Mixture Model using the Expectation-Maximization Algorithm from scratch in R. The model generated synthetic Gaussian datasets, estimated unknown parameters, performed clustering, and visualized the results without using external machine learning libraries.

The project demonstrates the practical implementation of probabilistic clustering techniques and provides a deeper understanding of how the EM algorithm iteratively improves model parameters to discover hidden structures within data. The results confirm that Gaussian Mixture Models are effective tools for clustering and density estimation problems.

References

1. Christopher M. Bishop, Pattern Recognition and Machine Learning.
2. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, The Elements of Statistical Learning.
3. Gaussian Mixture Models Documentation.
4. Expectation-Maximization Algorithm Research Papers.
5. R Programming Documentation.